instrumentation.py

`src/moe_pruning/model/instrumentation.py · 85 lines · module moe_pruning.model.instrumentation`

What it does

Generic checks and helpers built on top of the adapter's instrumentation hooks.

How it works

- `gate_consistency_error()` compares the layer's real output with the sum of gate times un-gated expert output. A near-zero error proves that the gate is applied exactly once and that outputs map to the right tokens.
- `gate_consistency_tolerance()` picks the tolerance for the model dtype (bfloat16/float16 have far fewer digits).
- `RecordCollector` keeps records in memory (for probes and tests).
- `assert_finite()` / `NonFiniteError` stop a run with an explanation when float16 overflows (inf/NaN).

Defined in this file

Extracted from the source code (classes, public functions and methods).

Name	Description
`reconstruct_moe_output(record)`	Recompute sum_k gate_k * E_k(x) from the recorded routing and expert outputs.
`gate_consistency_error(record)`	Compare the layer's real output with sum_k g_k * E_k(x).
`class RecordCollector`	Callback that keeps records in memory (for small probes and tests only).
`.__call__(record)`	Store one record (moved to CPU unless to_cpu=False).
`.by_layer()`	Records grouped by layer id.
`gate_consistency_tolerance(dtype)`	Relative tolerance for `gate_consistency_error` at the model's precision.
`class NonFiniteError`	Raised when inf/NaN values appear (typically float16 overflow).
`assert_finite(tensor, what, dtype)`	Raise a clear error on inf/NaN (typical for float16 overflow, e.g. bf16-trained models on T4).

Connections

Uses: `moe_pruning.model.architecture`

Used by: `scripts/collect_stats.py`, `scripts/inspect_model.py`, `scripts/replace_experts.py`, `src/moe_pruning/evaluation/evaluator.py`